Modular Arithmetic & $\mathbb{Z}/m\mathbb{Z}$

Idea: arithmetic w/remainders

Quotient-Remainder Thm:

If $a, m \in \mathbb{Z}$ & $m > 0$, then $\exists$ unique q & $r \in \mathbb{Z}$

st: $a = mq + \underbrace{r}_{\text{rem}(a,m)}$.

DEF 1.

If $a, b \in \mathbb{Z}$ st. $\text{rem}(a,m) = \text{rem}(b,m)$,

then we say <u>a is congruent to b modulo m</u>

measure

mod

We write $a \equiv b \bmod m$ OR $a \equiv_m b$

$3 \equiv 7 \bmod 4$

$3 \equiv_4 7$

$\rightarrow$ $3 = 0 \cdot 4 + 3$ $= \text{rem}(3,4)$

$\|$

$7 = 1 \cdot 4 + 3$ $= \text{rem}(7,4)$

$\boxed{\text{Thm. 1}}$

Congruent mod m is in an equivalence relation.

We can treat $\equiv_m$ <u>"sorta like" an</u> equality

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Thm. 1

Congruent mod m is in an equivalence relation.

We can treat $\equiv_m$ "sorta like" an equality

$a \equiv_m a$, $a \equiv_m b \Rightarrow b \equiv_m a$, $a \equiv_m b$, $b \equiv_c m \Rightarrow a \equiv_m c$

Thm. 2

$a \equiv \mathrm{Rem}(a, m) \bmod m$.

Pf

Write r for $\mathrm{Rem}(a, m)$.

quotient rem. thm.

Note $0 \le r < m$ by def. of r. Lemma 3 shows $\mathrm{Rem}(r, m) = r = \mathrm{Rem}(a, m)$.

By def. 1, $a \equiv r \equiv \mathrm{Rem}(a, m)$. $\square$

Lem. 3

If $0 \le m$, then $\mathrm{Rem}(c, m) = c$.

Pf

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Lem. 3

If $0 \leq m$, then $\text{rem}(c, m) = c$.

Pf

We have $c = qm + r$ where $0 \leq r < m$ for unique q, r. We also have $c = 0 \cdot m + c$ where $0 \leq c < m$.

From uniqueness, $r = \text{rem}(c, m) = c$. $\square$

Thm. 4

If $a, b < m$, then $a \equiv b \mod m$ iff $a = b$.

Corollary 5 reps. of equivalence classes

Suppose $a, m \in \mathbb{Z}$, $m > 0$.

$\exists$ exactly one $b \in \{0, 1, \ldots, m-1\}$ st. $a \equiv b \mod m$.

ex) $5000000 \equiv x \mod 11$ with $0 \leq x \leq 11$ has exactly one sol.

$x = 5$

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Thm. 6

$a \equiv b \bmod m \iff m \mid (a-b)$

Pf

Suppose $a \equiv b \bmod m$.

Let $r = \mathrm{Rem}(a, m) = \mathrm{Rem}(b, m)$.

Thus $a = q_1(m) + r$ $\quad b = q_2(m) + r$ for some $q_1, q_2 \in \mathbb{Z}$.

Hence $a - b = (q_1(m) + r) - (q_2(m) + r)$

$= m(q_1 - q_2)$.

Thus $m \mid (a-b)$.

Conversely, suppose $m \mid (a-b)$.

So $a - b = mk$.

$0 \le r < m$

Let $r = \mathrm{Rem}(b, m)$ so $b = q(m) + r$ for some $q \in \mathbb{Z}$. Thus $a = a \cdot b + b = mk + (q(m) + r)$

$= m(k+q) + r$

By uniqueness of Q-R Thm, the $\mathrm{Rem}(a, m) = r$.

Thus $a \equiv b \bmod m$. $\square$